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Traffic Sensors Selection for Complete Link Flow Observability through Simulated Annealing

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Abstract: Sensor selection is a key issue in sensor network design. Due to limited equipment and maintenance costs, and to minimize processing and communication overhead, only a limited number of sensors can reasonably be used, and they must also be reasonably placed. Indeed, the selection of an appropriate number of physical sensors is critical to ensure reliable estimates when monitoring large-scale systems such as traffic flows, water or gas transport infrastructures, or underwater wireless sensor networks. Regarding traffic systems, the full link flow observability problem consists of selecting the minimum number of traffic sensors to be installed from a given larger set (Salari, 2019). In particular, it involves selecting a subset of p, possibly redundant, sensors from a larger set of $n \gg p$ potential sensors in order to preserve the structural observability property of the entire traffic network. To solve this problem, the classical concept of system observability is exploited as a criterion for sensor selection. In this paper, we refer to the design of a simulated annealing heuristic to approximately solve the problem. The resulting subset of sensors is then used to design a Luenberger observer to properly estimate the system state. Some numerical simulations demonstrate the effectiveness of the proposed approach.

Keywords: Sensor selection, sensor network, simulated annealing, observability, integer optimization.

1. INTRODUCTION

In recent years, the smart cities paradigm has attracted worldwide attention, and several studies and research programs have been conducted to demonstrate the positive impact of smart technologies to improve the citizen well-being (Monzon, 2015; Harmon, 2015; Gagliardi, 2018). From a technological point of view, the smart cities paradigm requires the deployment of advanced devices and sensors (e.g. motion sensors, weather sensors, environmental sensors, etc.) to collect a large amount of data useful for specific applications. As an example, the deployment of such a large amount of sensors required to collect all the data necessary to monitor traffic flows in each sector of the road network of interest, is inappropriate for many practical reasons. In practice, only a limited number of sensors can be deployed due to economic and environmental constraints, and appropriate observers must be used to estimate the missing data. In view of this, the minimum sensor selection problem plays a key role.

Several approaches aimed at solving the above sensor selection problem have been reported in the literature. One of the most important contributions to this topic is (Georges, 1995), in which the optimal placement of both sensors and actuators for linear and non-linear dynamical systems is considered based on observability and controllability indices. Moreover, a solution to the sensor selection problem that maintains the structural observability for Discrete-Event systems modeled by Petri nets is presented in (Yu, 2007) whilst in(Joshi, 2009), a heuristic method based on convex optimization is described to approximately solve the problem of selecting a subset of sensors from a given set of possible sensors.

Various approaches have been developed for sensor placement problems in water networks (including integer programming models (Lee, 1992; Watson, 2004), quantum computation (Speziali, 2021), combinatorial heuristics and meta-heuristics (Kumar, 1999; Ostfeld, 2004)). On the other hand, the problem of placing measurement units in distribution networks has been successfully studied using Genetic Algorithms (GA) (Muller, 2016), Simulated Annealing (SA) algorithm (Yuan, 2014) and Evolutionary Algorithm (EA) (Prasad, 2018). Moreover, the sensor selection task has been successfully applied to various problems: e.g. refers to (Fu, 2012) for dynamical systems application, (Casavola, 2022) for quality of service specification, and (Schizas, 2013) for source-informative sensor identification.

This paper moves from the results presented in (Varotto, 2019), where the observability-based sensor placement problem is addressed through three observability metrics related to the rank, the trace, and the condition number of the Observability Gramian and solved as an integer nonlinear programming selection problem. More in detail, with reference to the results presented in (Gagliardi, 2020),

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in this work we address the specific problem of placing a fixed number of sensors in an urban street network for smart lighting dimming purposes. In this respect, the focus is to design an observer that, by exploiting a limited number of sensors (e.g. camera or motion sensors) properly placed in some sections (links) of the urban street network, is able to reconstruct the traffic flow which is important for determining the optimal lighting profile for all the street lights in the urban area. According to this specification, a traffic model for the urban area to be monitored is derived and the sensor selection metric is defined. Then, the sensor selection problem is formulated as a constrained optimization problem and solved using a simulated annealing heuristic (Gendreau, 2010). The resulting sensors can be used to design a suitable state observer. Here, a simple Luenberger state observer is considered mainly for assessment purposes. A more suitable solution, which also considers aspects of reconfiguration due to sensor faults, is presented in the companion paper (Gagliardi, 2023). Finally, numerical simulations are performed to validate the scheme and show the effectiveness of the proposed solution. A comparison with the sensor selection approach based on the Genetic Algorithms presented in (Varotto, 2019) is also reported.

The paper is organized as follows. After a mathematical preliminary section, in Section3 the traffic network and its model are provided. Section 4 describes the sensor selection problem, presents the accounted observability metrics and provides details on the Simulated Annealing heuristic used. Section 6 reports simulations for a case study that proves the effectiveness of the proposed sensor selection strategy. Finally, some conclusions end the paper.

2. PRELIMINARIES

In this paper, we consider a simplified traffic model described by the following discrete-time Linear Time-Invariant (LTI) state-space representation

$$x(t+1) = Ax(t) + Bu(t), \quad x(0) = x_0 \tag{1}$$

$$y(t) = Cx(t) + Du(t) \tag{2}$$

where $x(t) \in \mathbb{R}^n$, $u(t) \in \mathbb{R}^m$ and $y(t) \in \mathbb{R}^p$ are respectively the system state, input and output vectors, and the system matrices A, B, C and D are of appropriates dimensions.

One of the main structural properties of a dynamical system is its observability, i.e., the ability to estimate the initial state $x(t_0)$ from given data $(u(t), y(t))$, $\forall t \geq t_0$ and $\forall t_0 \geq 0$. This property holds if and only if the observability matrix

$$\mathcal{O} = \left[\, C^T,\, (CA)^T,\, \ldots, (CA^{n-1})^T \,\right]^T \tag{3}$$

has full $rank(\mathcal{O}) = n$. Notice that (3) is not useful for solving the sensor selection problem because it furnishes only a binary "on/off" indication and doesn't give any information on the quality of the Observability property. In order to consider a more suitable Observability performance criterion, it is possible to use the condition number of W (Varotto, 2019)

$$\mathcal{K}(W) = \frac{\overline{\sigma}(W)}{\underline{\sigma}(W)} \tag{4}$$

where

- W is the Observability Gramian defined as

$$W = \sum_{k=1}^{n-1} (A^T)^k C^T C A^k \tag{5}$$

- $\overline{\sigma}(W)$ and $\underline{\sigma}(W)$ are respectively the largest and the smallest singular values of the Observability Gramian.

3. SYSTEM DESCRIPTION AND PROBLEM STATEMENT

A traffic network can be considered as a wireless sensor network (WSN) in which a group of sensors (e.g. cameras, radar, motion sensors, etc.) is responsible for providing information about vehicle traffic in each monitored sector of the road network. Formally, a traffic network can be efficiently described by considering the compartmental modeling approach (Walter, 1998). The structure of the compartmental model is not conceptually complex: the system is partitioned into homogeneous compartments and the flows among them are traced. Figure 1 represents the structural aspect of the compartmental system, which can be viewed as a direct graph where the compartments are the vertices and the flows among them are the arcs. Referring to the definition of the road network model, we specify the number of vehicles within the i-th compartment as $x_i(t)$ and assume that the following condition holds for each compartment:

$$x_i(t) \geq 0, \forall t \geq 0 \tag{6}$$

The flow from the i-th to the j-th compartment is considered proportional to $x_i(t)$ according to a non-negative constant splitting ratio a_{ji} as:

$$f_{ji} = a_{ji} x_i(t), \quad j \neq i, \quad a_{ji} \geq 0 \tag{7}$$

In addition, it is necessary to emphasize the presence of compartments that are not considered by the model and conventionally considered as the unique *virtual* compartment 0. The latter is called the *external world* and the flow from it to the i-th compartment plays the role of system input. On the other hand, the flow from the i-th compartment to the external world is modeled as $f_{0i} = a_{0i} x_i(t)$.

Given a territory of interest $(\mathcal{T})$, it is possible to define the set of all road sectors contained therein as $\tilde{\mathcal{R}} = \{R_1, \ldots, R_M\}$. It is important to emphasise that not all road sectors belonging to $\mathcal{T}$ are equally important. As an example, the online vector map databases (e.g., OpenStreetMap - OSM)[1] ranks the roads into several *categories*. Here, seven road categories are considered where the least important is ranked $c_i = 1$, while the most important is ranked $c_i = 7$. In addition, road sectors considered of low importance in traffic monitoring applications (e.g. cul-de-sacs) can be removed from the initial set $\tilde{\mathcal{R}}$. In this way, the initial set $\tilde{\mathcal{R}}$ can be reduced to the set of principal roads $\mathcal{R} = \{R_1, \ldots, R_n\} \in \tilde{\mathcal{R}},\ n \leq M$. For each road sector $R_i \in \mathcal{R}$:

(1) the category of importance is obtained from the online vector maps OSM;
(2) we indicate as $\mathcal{P}$ the set of road sectors connected to the external world;

[1] `https://www.openstreetmap.org`

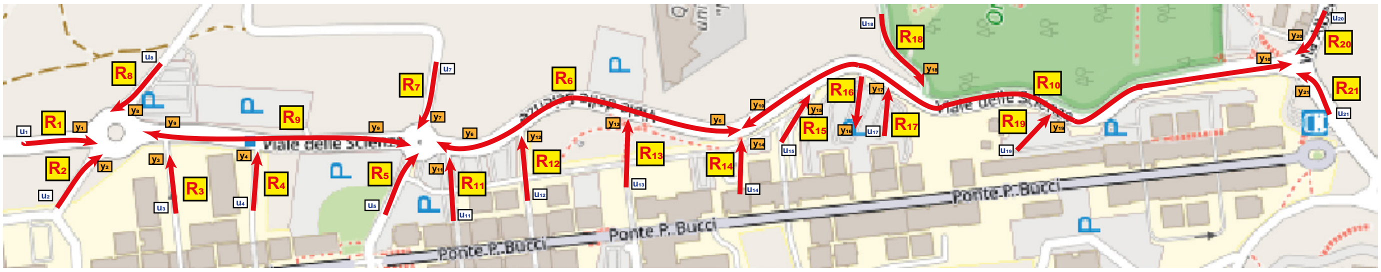

Fig. 1. Example of traffic compartmental system - Road schematic around University of Calabria. R_i indicate 21 potential sensor locations, u_i and y_i the cars, respectively, entering and leaving the i-th link.

(3) we define the matrix G whose entries are defined as:

$$g_{ij} = \begin{cases} 1 \; if \quad R_i \to R_j \\ 0 \; otherwise \end{cases} \tag{8}$$

where $R_i \to R_j$ means that vehicles can go from road R_i to road R_j. Note that, in general, G is not symmetric, due to the presence of possible one-way roads.

Then, the compartmental road network can be modeled as an LTI system (1)-(2) (Varotto, 2019), where:

- the state $x(t) \in \mathbb{R}^n$ is such that $x_i(t)$ represents the actual traffic flow (number of vehicles per unit time) on the road sector R_i;
- the elements of the state matrix $A \in \mathbb{R}^{n\times n}$ are the splitting ratios defined according to the roads importance rank as:

$$a_{ji,j\neq i}=\begin{cases} \dfrac{c_j}{\sum_{k:R_i\to R_k} c_k + c_i} & if \;\; R_i \to R_j \;\&\; R_i \notin \mathcal{P} \\ \dfrac{c_j}{\sum_{k:R_i\to R_k} c_k + 2c_i} & if \;\; R_i \to R_j \;\&\; R_i \in \mathcal{P} \\ 0 & otherwise \end{cases} \tag{9}$$

$$a_{ii,j\neq i} = 1 - \left(a_{0i} + \sum_{j\neq i} a_{ji} \right) \tag{10}$$

- The system input is chosen so that the i-th entry represents the traffic flow from the external world to road sector R_i and $B \in \mathbb{R}^n$;
- $y(t) \in \mathbb{R}^n$ is such that $y_i(t)$ represents the number of vehicles leaving the road sector R_i.

Eq. (9) means that the flow from R_i to R_j is proportional to the importance of R_j, normalized w.r.t. the importance of all other roads connected to R_i (and R_i itself). In addition, if $R_i \in \mathcal{P}$, we must consider the flow toward the external world, whose importance score is usually equated with R_i itself.

Problem Statement - Let $\mathcal{T}$ be a territory of interest to be monitored and denote by $\mathcal{R}$ the set of principal roads within $\mathcal{T}$. Consider the road network model (1),(2),(9),(10) and the condition number (4) as a measure of Observability. Denote by $\mathcal{S}$ the set of all n possible sensor placements, where each element $s_i \in \mathcal{S}$ represents the i-th sensor position. The problem at hand is to design a sensor selection procedure that is able to select a minimum subset of p sensors from a set of n potential sensor locations, such that the performance criterion (4) is optimized and the road network system Observability is guaranteed.

4. OBSERVABILITY BASED SENSOR SELECTION

Given the road network model (1),(9),(10), the goal of the sensor selection procedure is to determine a subset $\mathcal{J} \subseteq \mathcal{S}$ of p sensors from the set of n potential sensors such that a measure of Observability is optimized. The observability-based sensor selection problem can therefore be solved in terms of the Observability Gramian and can be formulated as a solution to the following mixed-integer constrained optimization problem (Hinson, 2014)

$$\begin{aligned} &\min_{\beta} \mathcal{K}[W(\beta)] \\ s.t. \; &\sum_{i=1}^{n} \beta_i = p \\ &\beta_i \in \{0,1\}\,, \;\; i = \ldots, n \end{aligned} \tag{11}$$

where $\mathcal{K}[\cdot]$ is a convex scalar measure of the Observability Gramian (i.e. the condition number) (Hinson, 2014) and the budget constraint (11) imposes that exactly p sensors must be selected. Moreover, it must be highlighted that for the sensors selection problem at the hand:

- with reference to the model (1), the following output equation is considered

$$y^*(t) = C(\mathcal{J})x(t) \tag{12}$$

where $C(\mathcal{J}) \in \mathbb{R}^{p\times n}$ indicates the road sectors where the sensors must be placed. It has p rows extracted from the $n \times n$ identity matrix according to the set of indices $\mathcal{J} = \{s_1, \ldots, s_p\}$;
- the Observability Gramian (5) is reformulated as

$$W(\beta) = \sum_{i=1}^{n} \beta_i \sum_{k=1}^{n-1} (A^T)^k C(\mathcal{J})^T C(\mathcal{J}) A^k \tag{13}$$

where β_i, $i, \ldots, n$, are binary "‘activation variables"’ whose meaning can be summarized by the function

$$\Lambda(\mathcal{J}) = \begin{cases} \beta_i = 1, \text{the sensor } s_i \text{ is selected} \\ \beta_i = 0, \text{the sensor } s_i \text{ is NOT selected} \end{cases} \tag{14}$$

It is important to note that the optimization problem (11) has a combinatorial nature and consists of finding a finite set of sensor locations that satisfy given conditions. A possible solution to this problem is exhaustive search (the brute-force approach), which consists of systematically enumerating all possible candidates for the solution and checking whether each candidate satisfies the problem's statement. However, the exhaustive search is NP-hard (Stetco, 2019) and testing all combinations becomes intractable for large networks (Olshevsky, 2014). For this reason, sub-optimal strategies, such as the heuristic procedure here proposed, can be used to solve (11) and obtain an appropriate set of sensors $\mathcal{S}^* \subseteq S$ with $|\mathcal{S}^*| = p$. We emphasise that the exhaustive search method allows one to find a global optimum $\mathcal{S}^{opt}$. In contrast, heuristic methods do not always guarantee to achieve the optimal selection $\mathcal{S}^* = \mathcal{S}^{opt}$. Moreover, because the optimization problem (11) is a nonlinear and non-convex mixed-integer programming problem (Varotto, 2019), we need to use a global optimization solver. Among many, we have investigated here the ability of *Simulated Annealing* (SA) and *Genetic Algorithm* search heuristics to find a suitable solution. These solutions are then compared to the solution of the brute-force enumeration approach. Only the details of the SA algorithm are presented for brevity, but the achieved results are compared for all three methods.

5. SIMULATED ANNEALING TECHNIQUES

Simulated Annealing (SA) is a local search algorithm that searches for a global optimum in the large search space of an optimization problem, especially of use when the search space is discrete. To formally describe the SA algorithm we introduce the following notation:

- the *solution space* Ω (i.e., the set of all possible solutions):
$$\Omega = \{\mathcal{J} \subseteq \mathcal{S} : rank(W(\beta)) = n;\ \ \beta = \Lambda(\mathcal{J})\} \quad (15)$$
- the objective function $\mathcal{K}[W(\beta)]$ defined on the solution space.

The goal is to find a global minimum, $\mathcal{S}^*$ i.e., $\mathcal{S}^* \subseteq \Omega$ such that $\mathcal{K}[W(\beta^*)] \le \mathcal{K}[W(\beta)]$ for all $\mathcal{J} \in \Omega$. Let's define $N(\mathcal{J})$ to be the neighbourhood set for $\mathcal{J}$. Therefore, associated with every solution, $\mathcal{J} \in \Omega$, there are neighbouring solutions, $\mathcal{J}' \in N(\mathcal{J})$, that can be reached in a single iteration of a local search algorithm. Simulated Annealing starts with an initial solution $\mathcal{J} \in \Omega$. A neighbouring candidate solution $\mathcal{J}' \in N(\mathcal{J})$ is then generated (either randomly or using some pre-specified rule). Simulated Annealing is based on the Metropolis acceptance criterion (Gendreau, 2010), which models how a thermodynamic system moves from the current solution (state) $\mathcal{J} \in \Omega$ to a candidate solution $\mathcal{J}' \in N(\mathcal{J})$, in which the energy content is being minimized. The candidate solution $\mathcal{J}'$ is accepted as the current solution on the basis of the following acceptance probability:

$$P(\mathcal{J}') = \begin{cases} exp(-(\Delta/t_k)) & if\ \Delta > 0 \\ 1 & otherwise \end{cases} \quad (16)$$

where $\Delta = \mathcal{K}[W(\beta')] - \mathcal{K}[W(\beta)]$ and t_k is the temperature parameter at iteration k, such that

$$t_k := \left\{ t_k > 0, \forall k, \lim_{k\to\infty} t_k = 0 \right\} \quad (17)$$

SIMULATED ANNEALING PSEUDO-CODE
INPUT: p
OUTPUT: $\mathcal{S}^*$
OBJ. FUNCTION TO BE MINIMIZED: $\mathcal{K}[W(\beta)]$

Select an initial solution $\mathcal{J} \in \Omega$
Define $\beta = \Lambda(\mathcal{J})$
Select the temperature change counter $k = 0$
Select a temperature cooling schedule, t_k
Select an initial Temperature $T = t_0 \ge 0$
Select a repetition schedule, M_k, that defines the number of iterations executed at each temperature, t_k
do
    Set repetition counter $m = 0$
    do
        Generate a solution $\mathcal{J}' \in N(\mathcal{J})$
        Define $\beta' = \Lambda(\mathcal{J}')$
        Compute $\Delta = \mathcal{K}[W(\beta')] - \mathcal{K}[W(\beta)]$
        if $\Delta \le 0$ **then**
            $\mathcal{J} = \mathcal{J}'$
            $\beta = \beta'$
        end if
        if $\Delta_{\mathcal{J},\mathcal{J}'} > 0$ **then**
            $\mathcal{J} = \mathcal{J}'$
            $\beta = \beta'$
            with probability $exp(-\Delta/t_k)$
        end if
        $m \leftarrow m + 1$
    while $m == M_k$
    $k \leftarrow k + 1$
    $T \leftarrow t_k$
while $|\mathcal{J}| == p$
$\mathcal{S}^* = \mathcal{J}$

6. SIMULATIONS

The aim of this section is to show the applicability of the proposed Simulated Annealing based sensor selection method by considering a traffic network composed of $\mathcal{R} = \{R_1, \dots, R_{21}\}$ road sections (see Figure 1). The first step was the determination of the road network model (1). A solution to the sensor selection problem was obtained by properly using the built-in MATLAB functions `simulannealbnd` for integer/discrete optimization. The problem with $n = 21$ is still tractable and we were able to find the minimal subset $\mathcal{S}^*_{sa}$ of $\mathcal{S}$ ensuring the full Observability of the system (1)-(12), that is

$$rank(O(A, C(\mathcal{S}^*_{sa}))) = 21 \quad (18)$$

In particular, the following results were obtained as a solution to the sensor selection problem

- Computation time: $94.5[sec]$
- Cardinality $|S^*_{sa}| = 6$;
- $\mathcal{K}(W) = 4.1102 * 10^3$.
- $\mathcal{S}^*_{sa} = [s_3, s_{13}, s_{15}, s_{17}, s_{19}, s_{20}]$.

Further tests were performed to investigate whether the solution provided by the Simulated Annealing procedure represents the minimum sensor set that allows the system Observability. In this regard, two types of procedures were considered: 1) Genetic Algorithm sensor selection procedure as in (Varotto, 2019); 2) Brute-Force (exhaustive) search. Using the Genetic Algorithm procedure, we were able to obtain the following results: *1)* Computation time: $89.6[sec]$; *2)* Cardinality $\left|S^*_{ga}\right| = 7$; *3)* $\mathcal{K}(W) = 7.6840 * 10^4$; *4)* $\mathcal{S}^*_{ga} = [s_3, s_5, s_{12}, s_{15}, s_{17}, s_{19}, s_{20}]$. On the other side, the exhaustive search provide the results reported in Table 1.

It can be noted that: *1)* the exhaustive search return 12 sets of sensors allowing system Observability; *2)* the obtained solutions are characterized by different values of the con-

#	Sensor sets	$\mathcal{K}(W)$
1	$[s_3, s_{13}, s_{16}, s_{17}, s_{19}, s_{20}]$	$4.1208 * 10^3$
2	$[s_3, s_{12}, s_{16}, s_{17}, s_{19}, s_{20}]$	$4.1509 * 10^3$
3	$[s_3, s_{13}, s_{15}, s_{17}, s_{19}, s_{20}]$	$4.1102 * 10^3$
4	$[s_3, s_{12}, s_{15}, s_{17}, s_{19}, s_{20}]$	$4.1509 * 10^3$
5	$[s_3, s_{13}, s_{15}, s_{16}, s_{19}, s_{20}]$	$4.1719 * 10^3$
6	$[s_3, s_{12}, s_{15}, s_{16}, s_{19}, s_{20}]$	$4.1891 * 10^3$
7	$[s_3, s_{13}, s_{16}, s_{17}, s_{18}, s_{20}]$	$4.2209 * 10^3$
8	$[s_3, s_{12}, s_{16}, s_{17}, s_{18}, s_{20}]$	$4.3519 * 10^3$
9	$[s_3, s_{13}, s_{15}, s_{17}, s_{18}, s_{20}]$	$4.1369 * 10^3$
10	$[s_3, s_{12}, s_{15}, s_{17}, s_{18}, s_{20}]$	$4.1869 * 10^3$
11	$[s_3, s_{13}, s_{15}, s_{16}, s_{18}, s_{20}]$	$4.4511 * 10^3$
12	$[s_3, s_{12}, s_{15}, s_{16}, s_{18}, s_{20}]$	$4.1519 * 10^3$

Table 1. Sensor sets computed via exhaustive search.

dition number (4); *3)* the sensor set number three is the same as the solution obtained by the Simulated Annealing procedure and it is the solution with the minimum value of the condition number $\mathcal{K}(W)$. Based on the obtained results, it can be stated that both SA and GA procedures are able to find a sensor set allowing system Observability. The main difference between the two methods lies in the cardinality of the sensor set and the computation time. It can be observed that the Simulated Annealing method is able to find a smaller sensor set than the Genetic Algorithm, but requires a slightly longer computation time (about +5% compared to the Genetic Algorithm). With respect to our case study, this difference can be considered acceptable, but for larger application scenarios where the number of sensors to be considered is very large, the choice of an efficient local search heuristic can be crucial.

6.1 State reconstruction error

Two state observers in the Luenberger form were designed according to the sensor sets (S^*_{sa}) and $\left(S^*_{ga}\right)$ respectively, and their state reconstruction errors were compared. The observer's matrix gains were designed in an optimal H_2 sense; in this way, the observer's expected mean-square state reconstruction error (MSE) in response to a white noise input excitation is minimized. In this respect, we also assume that the measurements provided by the sensors are subject to the exogenous disturbance $v(t) = WN(0, \sigma)$. Figures 2-3 show the obtained results. Figure 2 focuses in particular on the evolution of system state x_1. In this Figure, the true state x_1 is represented by the blue line and it can be observed that the state estimate ($\hat{x}_{1,sa}$ - green dashed line), obtained by using the set (S^*_{sa}) is more accurate than the state estimate ($\hat{x}_{1,ga}$ - red dashed line) obtained by using the set $\left(S^*_{ga}\right)$. Similar results are obtained for all system state estimates (x_i, $i = 2, \ldots, 21$). On the other hand, Figure 3 shows the mean squared reconstruction errors for all states (x_i) computed over the entire simulation interval:

$$e_k = \frac{\sum_{i=1}^{t_{final}} \left(x(t) - \hat{x}_k(t)\right)^2}{t_{final}}, \quad k = \{sa, ga\} \qquad (19)$$

where $\hat{x}_{sa}$ and $\hat{x}_{ga}$ are the estimates of the system state obtained by using the sensors set(S^*_{sa}) and $\left(S^*_{ga}\right)$ respectively. From the results presented, it can be highlighted that the observer designed for the sensor set (S^*_{sa}) performs better than the observer designed for the sensor set $\left(S^*_{ga}\right)$.

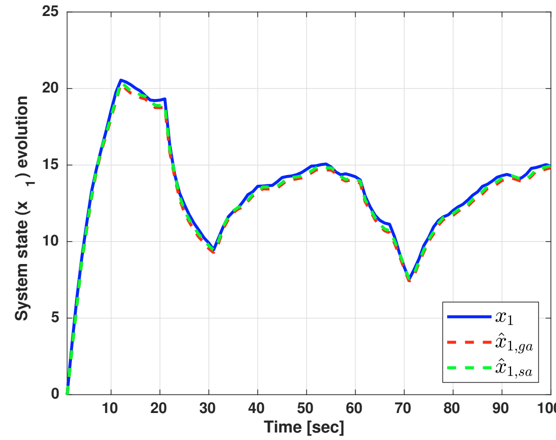

Fig. 2. State observer performance evaluation - system state (x_1) estimation: real (blue line), Genetic Algorithm (red dashed line) and Simulated Annealing (green dashed line) based estimations.

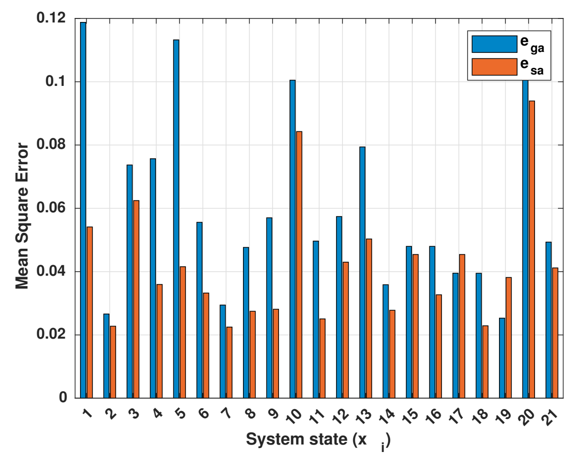

Fig. 3. State observer performance evaluation - mean squares error for the Genetic Algorithm (blue bar) and the Simulated Annealing (red) procedures based observers.

7. CONCLUSIONS

In this paper, a sensor selection problem for traffic monitoring in smart road lighting systems was formulated. Based on a compartmental network traffic model, the problem can be transformed into a mixed-integer nonlinear programming problem. A global minimum was found by the Simulated Annealing heuristic by minimizing the condition number of the Observability Gramian under a budget constraint on the maximum number of sensors to be used, which were selected from a larger set. Numerical simulations were used to validate the approach, whose solutions were compared with the solutions of other heuristics based on Genetic Algorithms and exhaustive search. The obtained results show that the proposed approach is quite simple to use and generates a minimal set of sensors that ensure good Observability properties and state reconstruction performance.

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REFERENCES

Chen, C.,-T., *Linear System Theory and Design.* Philadelphia, PA, USA: Saunders College Publishing, 3rd Ed., 1998.

Casavola, A., Franzé, G., Gagliardi, G. and Tedesco, F., "Improving Lighting Efficiency for Traffic Road Networks: A Reputation Mechanism Based Approach," in *IEEE Transactions on Control of Network Systems*, 2022, doi: 10.1109/TCNS.2022.3182026.

Fu, Y., Ling, Q. and Tian, Z., "Distributed sensor allocation for multi-target tracking in wireless sensor networks," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 48, no. 4, pp. 3538-3553, 2012.

Gagliardi, G., Casavola, A., Lupia, M., Cario, G., Tedesco, F., Lo Scudo, F., Gaccio, F.C., Augimeri, A., "A smart city adaptive lighting system", in *3rd International Conference on Fog and Mobile Edge Computing, FMEC 2018*, Barcelona, 2018.

Gagliardi, G., Lupia, M., Cario, G., Cicchello Gaccio, F. Lo Scudo, F., Tedesco, F. and Casavola, A., "Advanced Adaptive Street Lighting Systems for Smart Cities", *Smart Cities*, Special Issue on "Feature Papers for Smart Cities", Vol. 3, N. 4, pp. 1495-1512, 2020.

Gagliardi, G., Casavola, A., D'Angelo, V., "Reliable Traffic Sensor Networks Via Fault-Tolerant Sensor Reconciliation Schemes", in *The 22nd World Congress of the International Federation of Automatic Control 2023*, Yokohama, JAPAN

Georges, D., "The use of observability and controllability gramians or functions for optimal sensor and actuator location in finite-dimensional systems," *Proceedings of 1995 34th IEEE Conference on Decision and Control*, 1995, pp. 3319-3324 vol.4.

Harmon,R., R., Castro-Leon, E., G. and S. Bhide, "Smart cities and the Internet of Things," *2015 Portland International Conference on Management of Engineering and Technology (PICMET)*, 2015, pp. 485-494.

Hinson, B., T., "Observability-based guidance and sensor placement," Ph.D. dissertation, Seattle, 2014.

Joshi, S. and Boyd, S., "Sensor Selection via Convex Optimization," in *IEEE Transactions on Signal Processing*, vol. 57, no. 2, pp. 451-462, Feb. 2009.

Lee, B. H., and Deininger, R. A., "Optimal locations of monitoring stations in water distribution system." *J. Environ. Eng.*, 118(1), pp. 4–16, 1992.

Liepins, G.E., Hilliard, M.R., *Genetic algorithms: Foundations and applications.* Ann. Oper. Res. 21, 31–58 (1989).

Monzon, A., "Smart cities concept and challenges: Bases for the assessment of smart city projects," *2015 International Conference on Smart Cities and Green ICT Systems (SMARTGREENS)*, 2015, pp. 1-11.

Muller, H., H. and Castro, C., A., "Genetic algorithm-based phasor measurement unit placement method considering observability and security criteria," *IET Generation Transmission & Distribution*, vol.10, no. 1, pp. 270-280, Jan. 2016.

Olshevsky, A., "Minimal Controllability Problems", in IEEE Transactions on Control of Network Systems, vol. 1, no. 3, pp. 249-258, Sept. 2014.

Ostfeld, A., and Salomons, E., "Optimal layout of early warning detection stations for water distribution systems security". *J. Water Resour. Plan. Manage.*, 130(5), 377–385, 2004.

S. Prasad and D. M. V. Kumar, "Trade-offs in PMU and IED deployment for active distribution state estimation using multi-objective evolutionary algorithm," *IEEE Transactions on Instrumentation and Measurement*, vol. 67, no. 6, pp. 1298-1307, Jun. 2018.

Gendreau, M., Potvin, J., -Y., (eds.), *Handbook of Metaheuristics*, International Series in Operations Research & Management Science 146, Springer Science+Business Media, LLC 2010

Salari, M., Kattan, L., Lam, W.H.K., Lo, H.P., Esfeh, M.A., "Optimization of traffic sensor location for complete link flow observability in traffic network considering sensor failure", *Transportation Research Part B: Methodological*, Volume 121, 2019, Pages 216-251.

Schizas, I., D., "Distributed Informative-Sensor Identification via Sparsity-Aware Matrix Decomposition," in *IEEE Transactions on Signal Processing*, vol. 61, no. 18, pp. 4610-4624, Sept.15, 2013.

S. Speziali et al., "Solving Sensor Placement Problems In Real Water Distribution Networks Using Adiabatic Quantum Computation," *2021 IEEE International Conference on Quantum Computing and Engineering (QCE)*, Broomfield, CO, USA, 2021, pp. 463-464.

Stetco, A., Dinmohammadi, F., Zhao, X., Robu, V., Flynn, D., Barnes, M., Keane, J., Nenadic, G., "Machine learning methods for wind turbine condition monitoring: A review", in *Renewable Energy*, Volume 133, 2019, Pages 620-635, ISSN 0960-1481.

Kumar, A., Kansal, M. L., and Arora, G., "Discussion of 'detecting accidental contaminations in municipal water networks' by A. Kessler, A. Ostfeld, and G. Sinai". *J. Water Resour. Plan. Manage.*, 125(4), 308–309, 1999.

Varotto, L., Zampieri, A. and Cenedese, A., "Street Sensors Set Selection through Road Network Modeling and Observability Measures", *2019 27th Mediterranean Conference on Control and Automation (MED)*, 2019, pp. 392-397.

Walter, G., G. and Contreras, M., *Compartmental Modeling with Networks*, 1st ed. Secaucus, NJ, USA: Birkhauser Boston, Inc., 1998.

Watson, J.P., Greenberg, H. J., and Hart, W. E. , "A multipleobjective analysis of sensor placement optimization in water networks", *Proc., World Water and Environment Resources Conf. ASCE*, 2004.

Yu Ru, Christoforos N. Hadjicostis, "Optimal sensor selection for structural observability in discrete event systems modeled by Petri Nets", *IFAC Proceedings Volumes*, Volume 40, Issue 6, 2007, Pages 223-228.

Yuan, P., Ai, Q. and Zhao, Y., "Research on multi-objective optimal PMU placement based on error analysis theory and improved GASA," *Proceedings of the CSEE*, vol. 34, no. 13, pp. 2178-2187, May. 2014.